

Per-Word Arabic *I'rāb* Generation: A Comparison Study with a Cross-Register Audit

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May 2026

Abstract

Arabic *i'rāb* (إعراب) generation produces structured grammatical analysis as natural-language prose. We report two findings that we believe should reframe the next research step on the task. First, **scale within Arabic-pretrained open-weight fine-tuning is teacher-bound**: three Arabic-pretrained models from 296M (AraT5v2-base) to 13B parameters (AceGPT-13B), trained on the same 77K-row Claude-Haiku-distilled corpus, reach statistically indistinguishable Gazelle performance — every pairwise McNemar $p=1.000$ at $n=134$ word judgments. The students inherit the teacher’s case-67% ceiling; on this benchmark, additional capacity has no signal to exploit. Second, **register transfer breaks the strongest system the most**. Moving from MSA news (Gazelle) to Quranic (a 5,007-word MASAQ subset), every LLM-trained system loses paired-significant role-F1 (≥ 14 pp); the closed-source frontier — Claude Sonnet 4.5 + RAG, our Gazelle headline at 32.1% fully-correct [24.6, 40.3] — shows the *largest* cross-register drop of any system tested (+61.7 pp★). A retrieval-pool-confound experiment rules out the obvious explanation: removing the MSA retrieval pool (Sonnet zero-shot) widens the drop to +67.0 pp★ rather than narrowing it. Within the resolution this benchmark allows, register and construction coverage — not parametric capacity — account for the remaining error. Alongside the comparison, §3.6 reports an interpretable structured-prediction rebuild of the original from-scratch baseline (AraT5v2 encoder + four classification heads + four soft symbolic-constraint families + deterministic prose template renderer) that lifts case from 32.8% to 55.2%, role-F1 from 3.8% to 36.9%, and *fully* from 2.2% to 18.7%, recovering most of the open-weight ceiling without sacrificing structured controllability. Two follow-on phases on the rebuilt baseline expose a clean structural finding. **Phase 1** (§3.6) adds seven auxiliary morphology heads trained jointly on UD Arabic-PADT; this lifts Gazelle role-F1 to 42.3% (+5.4 pp) and *fully* to 19.4% (+1.5 pp) without changing the *i'rāb* head architecture. **Phase 4a** (§3.6) expands the canonical role taxonomy from 25 to 34 labels; granularity-alone independently lifts role-F1 to 43.7% (+6.8 pp) and case to 56.7%. The 2×2 ablation (§3.6) surfaces an un-

expected result: granularity and morphology contribute almost the *same* role-F1 signal — combining them lifts role-F1 by only +5.7 pp, far below the +12.2 pp linear sum of their individual gains — so they are **partially substitutable, not additive**. Granularity wins on case + marker; morphology wins on the *fully* aggregate. **Phase 2** (§3.6) tests the natural follow-on — *hierarchical conditioning* — across three mechanisms (FiLM joint, additive joint, FiLM detached). All three fail the strict ship gate, but the pattern is informative: FiLM-joint regresses all four metrics, while FiLM-detached and additive-joint preserve case + marker + *fully* and degrade only role-F1 by 1–3 pp. *The same conditioning module*, with the only difference being whether *i'rāb*-side gradients flow back into the morph heads, produces qualitatively different outcomes. The bottleneck is **joint optimisation dynamics**, not the form of the conditioning interaction — the morph representation drifts under joint training, and the *i'rāb* heads chase that moving target. **Phase 3** (§3.6) tests the prediction this finding makes: the next productive lever is an *independent signal source*, not another rearrangement of the same supervision. Phase 3 feeds Stanza-parsed UD dependency edges (DEPREL, HEAD topology, governor’s POS) as static input augmentation to the *i'rāb* decoders — sidestepping the Phase 2 joint-dynamics issue because the dep signal is computed offline, not learned through a head. **Phase 3-A passes the soft ship gate**: case 56.7 (+3.0 vs Phase 1), role-F1 41.3 (−1.0), marker 44.8 (+3.8), *fully* 20.1 (+0.7). **Three of four metrics improve simultaneously** — the first architectural intervention since Phase 1 to do so. Phase 3-A becomes the new production checkpoint. Phase 1 remains a documented baseline; Phase 4a (taxonomy expansion) and Phase 2 (soft conditioning) remain opt-in.

1 Introduction

The *i'rāb* task takes an undiacritized Arabic sentence and produces per-word traditional grammatical analysis as Arabic prose. Each word receives four fields: part of speech, case (*rafʿ/naṣb/jarr/jazm/mabni*), syntactic role (~25 traditional labels), and marker phrase (~200 unique surface forms, e.g. على الظاهرة الضمة

آخره). The task matters for two practical reasons. First, every Arabic-as-a-second-language curriculum uses *i'rāb* as the canonical pedagogy framing for Arabic syntax. An automatic *i'rāb* tool would directly help language learning, religious-studies tooling, and corpus annotation. Second, the task probes fine-grained Arabic grammatical knowledge that POS accuracy or dependency UAS misses. The role + case + marker triple requires resolving subject/object selection, *iḍāfa* state, and exception (*istiṭnā*) constructions whose interactions are non-trivial.

The pain. The literature on Arabic *i'rāb* reads as two non-comparable threads. Classifier pipelines (CamelParser2.0, MADAMIRA, Farasa) report dependency / morphological accuracy on tag-level outputs. Frontier-LLM evaluations (e.g. Gazelle) report multiple-choice or string-similarity scores on prose. Two questions remain unanswered. (i) When the same system is asked to *generate* prose *i'rāb*, do extra parameters help, or does the training data ceiling dominate? (ii) Does a system that scores well on MSA news still score well on a different register (e.g. Quranic), or does the headline number hide a register-specific competence?

The gain. We answer both questions on a controlled testbed. A regex/FSM extractor parses every prediction into {case, role, marker}, and every system-vs-system delta carries a paired bootstrap CI and a McNemar exact p -value (the methodology is infrastructure for the findings, not a contribution in itself). Within this framework we evaluate eleven systems — one classifier-pipeline baseline (Stanza), one open-weight LLM peer (Qwen2.5-7B+RAG), four open-weight fine-tuned models from 296M to 13B trained on the same Haiku-distilled corpus, four closed-source LLM variants (Haiku and Sonnet, zero-shot and +RAG), and one routing hybrid (Mix A). We then re-evaluate the six trainable / closed systems on a held-out 5,007-word Quranic MASAQ subset.

Contributions.

1. **Teacher-bound scaling on *i'rāb* fine-tuning** (§3.2, Fig. 1). Three Arabic-pretrained open-weight models spanning 296M to 13B parameters, trained on a single Claude-Haiku-distilled corpus, reach statistically indistinguishable Gazelle performance (every pairwise Δ has McNemar $p=1.000$). The teacher's $\sim 67\%$ case ceiling is the binding constraint, not the student's capacity. We treat this as a constraint specific to Haiku-as-teacher and the Gazelle benchmark size, and we propose a Sonnet-redistillation experiment that would directly test whether scale begins to matter once the teacher ceiling lifts.

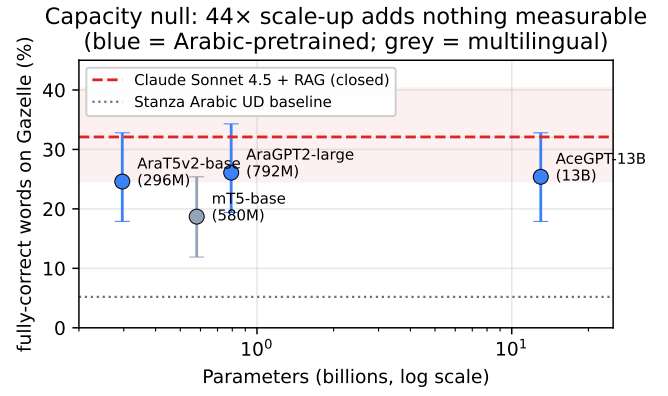


Figure 1: Gazelle *fully-correct* rate vs. open-weight parameter count (log scale). Three Arabic-pretrained models (296M, 792M, 13B; blue) train to statistically indistinguishable performance — every pairwise McNemar $p=1.000$. The closed-system frontier (Sonnet RAG, dashed red) sits 7–8 pp above. Multilingual mT5 (grey) trails by paired-significant margin. Scale within Arabic-pretrained open weights does not move the headline.

2. **Cross-register fragility, peaked at the strongest system** (§3.5). The closed-source frontier on MSA news (Sonnet RAG, +61.7 pp★ drop, retains 19% of MSA role-F1 on Quranic) is the system most disturbed by the register shift, not the most robust. A confound experiment rules out the natural “the retrieval pool is MSA” explanation: zero-shot Sonnet drops *more*, not less. Within the resolution of this benchmark, the remaining error is dominated by register and construction coverage rather than by model capacity, which reframes the next research step toward targeted annotation rather than larger models.

2 Experiments

2.1 Data

We use three categories of *i'rāb* data, distinguished by their level of human authorship.

Manual gold (held-out evaluation). *Gazelle* [5]: 30 hand-written MSA sentences ($n=134$ word judgments). *MASAQ* [12]: 624 Quranic verses ($n=5,007$ word judgments); the 999-word subset where the gold contains an extractable role label provides a matched cross-register surface. *Yarob* (linuxscout/yarob, GPL): 459 MSA *i'rāb* sentences in classical-grammar prose; we use these as the retrieval pool, not as evaluation.

LLM-distilled training corpus. We use Claude Haiku 4.5 as a teacher to generate per-word *i'rāb* for 5,000 PADT MSA news sentences via the Anthropic Messages Batches API. After JSON-schema validation, we obtain **77,534**

(word, sentence, irab) rows. This is the training corpus for every open-weight FT model below; no model trains on Gazelle or MASAQ.

Construction coverage. A regex over each Gazelle gold prose tags the dominant construction(s) per sentence using 11 traditional categories. The 30-sentence eval is dominated by short verbal sentences ($n=15$) and prepositional phrases ($n=8$); exception (*istithnā*) and *kāna* sisters appear in 2 sentences each ($n=9$ and $n=7$ words). Generalization claims are limited to these construction frequencies.

2.2 Models

We evaluate eleven systems on Gazelle.

Classifier-pipeline baseline. *Stanza Arabic* [10] with a deterministic UD $\rightarrow$ *i'rāb* stub (the published-prior-work peer).

Open-weight LLM peer. *Qwen2.5-7B-Instruct* + RAG ($k=5$, 4-bit local).

Open-weight FT models (same 77,534-row Haiku corpus, same prompt template):

- AraT5v2-base (296M, T5 enc-dec) [8] — full FT, 3 epochs.
- mT5-base (580M, T5 enc-dec) [13] — full FT, 3 epochs.
- AraGPT2-large (792M, GPT-2 decoder-only) [3] — LoRA $r=32$, 3 epochs.
- AceGPT-13B (13B, Llama-2 decoder-only) [6] — QLoRA 4-bit $r=32$, 1 epoch.

Closed-source LLMs. Claude Haiku 4.5 (zero-shot, +RAG $k=5$); Claude Sonnet 4.5 (zero-shot, +RAG $k=5$). RAG uses Jaccard retrieval over a 1,060-example pool (Yarob + a 601-row distilled prior) following the standard recipe [7].

Hybrid Mix A. The LLM-RAG output supplies case and role; the AraT5v2-base specialist supplies the marker phrase, called per word with the structured prompt [case= c] [role= r] $w \mid s \rightarrow$ marker.

A six-system subset (Stanza, mT5, AraT5v2, AraGPT2, AceGPT-13B, Sonnet RAG) also runs on MASAQ; the AceGPT-13B MASAQ run hit a 4 h SLURM timeout at 21% of MASAQ ($n=1,075$ words), and we report it with that caveat.

2.3 Setup

Structural metric. A regex/FSM extractor parses each generated *i'rāb* string into four fields: pos, case $\in \{\text{raf}^f, \text{naṣb}, \text{jarr}, \text{jazm}, \text{mabni}\}$, role (~ 25 labels), marker (~ 200 phrases). Macro role-F1 over a closed label set L is

$$\text{role-F1}_{\text{macro}} = \frac{1}{|L|} \sum_{\ell \in L} \frac{2P_{\ell}R_{\ell}}{P_{\ell} + R_{\ell}}, \quad (1)$$

where P_{ℓ}, R_{ℓ} are precision and recall for label ℓ . We additionally report **case-acc**, **marker-EM**,

well-formed, and the headline aggregate

$$\text{fully}(w) = \mathbf{1}[\text{case}(w)] \cdot \mathbf{1}[\text{role}(w)] \cdot \mathbf{1}[\text{marker}(w)]. \quad (2)$$

We avoid chrF and BLEU because partial-credit on string overlap masks morphological errors that we explicitly target.

Statistical inference. Every per-system metric carries a 95% percentile bootstrap CI ($B=1000$):

$$\text{CI}_{0.95}(\theta) = [\theta_{(0.025B)}^*, \theta_{(0.975B)}^*]. \quad (3)$$

For paired comparisons of two systems on the same n word judgments, we report the paired-bootstrap CI of $\Delta = \theta_A - \theta_B$ and McNemar's exact p -value on the discordant-pair count b/c :

$$p_{\text{McNemar}} = 2 \min \left(\sum_{k=0}^{\min(b,c)} \binom{b+c}{k} 2^{-(b+c)}, 0.5 \right), \quad (4)$$

computed in log-space for $n > 1000$. A Δ counts as significant ($\star$) only if its 95% paired CI excludes 0 and $p_{\text{McNemar}} < 0.05$. Cross-register comparisons (Gazelle subset $n=78$ vs. MASAQ subset $n=999$) use a two-sample bootstrap, since the surfaces share no items. The smallest detectable difference at $n=134$, $\alpha=0.05$, power 0.80 is approximately ± 7 pp on a binary metric.

Extractor audit. We perturb each Gazelle gold record in three ways (case-flip, role-flip, marker-mangle), producing 402 records, and re-score with the same extractor. Specificity reaches 100% on unperturbed controls (no false positives); sensitivity is 92.7% (marker), 88.2% (case), 60.0% (role). The 60% role floor reflects the closed taxonomy of ~ 25 traditional labels; we read role results with this caveat.

3 Results

3.1 Headline (Gazelle)

Table 1 shows the headline. Three findings follow.

3.2 Three null results from the comparison

(a) Open-weight scale is null on Gazelle. AraT5v2-base (296M), AraGPT2-large (792M), and AceGPT-13B (13B) — three architectures (T5 enc-dec, GPT-2 decoder-only, Llama-2 decoder-only), $44\times$ parameter range — train to statistically indistinguishable Gazelle performance. Every pairwise Δ has McNemar $p=1.000$ (e.g. AceGPT-13B vs. AraT5v2-base: case $\Delta +0.7$, marker $\Delta -0.7$, fully $\Delta +0.0$). Fig. 1 visualises the result.

Table 1: Headline comparison on Gazelle ($n=134$ word judgments). All numbers are percentages; brackets are 95% percentile bootstrap CIs ($B=1000$) shown for the headline metric *fully* only. Boldface = paired-best within column. The from-scratch decoder appears in Appendix A.

System	well	case	role-F1	marker	fully
Stanza Arabic (UD+stub)	59.7	35.1	10.3	13.4	5.2 [2.2, 9.7]
Qwen2.5-7B-Instruct+RAG	66.4	43.3	19.2	19.4	3.0 [0.0, 6.0]
mT5-base (580M) FT	79.9	61.9	31.6	32.8	18.7 [11.9, 25.4]
AraT5v2-base (296M) FT	79.9	65.7	54.8	44.0	24.6 [17.9, 32.8]
AraGPT2-large (792M) LoRA FT	79.9	64.9	54.6	43.3	26.1 [19.4, 34.3]
AceGPT-13B QLoRA FT (1 epoch)	79.1	66.4	55.5	43.3	25.4 [17.9, 32.8]
Claude Haiku 4.5 zero-shot	77.6	57.5	55.9	40.3	18.7 [11.9, 25.4]
Claude Haiku 4.5 + RAG	79.9	67.2	68.9	44.8	27.6 [20.1, 35.8]
Mix A (Sonnet RAG + AraT5v2 marker)	79.9	73.9	73.3	46.3	29.1 [21.6, 37.3]
Claude Sonnet 4.5 zero-shot	78.4	72.4	76.4	44.0	27.6 [20.1, 35.8]
Claude Sonnet 4.5 + RAG	79.9	73.9	74.7	50.0	32.1 [24.6, 40.3]

(b) Mix A’s per-word routing does not significantly improve over Sonnet RAG. Δ case +0.0 pp ($p=1.000$), Δ marker -3.7 pp ($p=0.125$), Δ fully -3.0 pp ($p=0.219$). The routing hypothesis — that case/role decompose from marker phrasing — does not hold here. Two factors plausibly explain why. The marker specialist trains on 8,815 marker pairs of which 7,172 (81%) are themselves Haiku-distilled; the specialist therefore learns “what Haiku would have said” rather than an independent gold style, and RAG already exposes Sonnet to that same demonstration distribution. Additionally, the Sonnet-base hybrid rebuild template truncates longer LLM role labels (e.g. إلى الجرح بحرف مجرور اسم) collapses to اسم (مجرور اسم), which slightly hurts role-F1. The result also holds on the Haiku base (Δ fully -1.5 pp, $p=0.791$); we report Mix A as a documented negative result on the routing decomposition at the present evaluation scale.

(c) Arabic-specific pretraining at 296M beats multilingual pretraining at 580M. AraT5v2-base – mT5-base: Δ marker +5.2 pp* ($p=0.039$), Δ fully +6.0 pp* ($p=0.021$). Same training corpus, same recipe, same architecture — only the pre-training corpus differs.

The closed-system advantage is real but bounded: Sonnet RAG beats AraT5v2-base by Δ case +8.2 pp* and Δ fully +7.5 pp*. Open-weight scale alone does not narrow this gap.

3.3 Per-construction failures

Fig. 2 surfaces two cross-system structural failures: EXCEPTION (0/9 across every evaluated system) and KANA SISTERS (0/7 across four of five). Both failures hold for the trained AraT5v2-base and AceGPT-13B as well as for every Claude variant. We read these as construction-frequency artefacts rather than as a Claude-specific stylistic gap. Exception (*istithnā*) requires recognising that the post-marker noun (after إله/عدا/سوى) takes case based on whether the exception is positive or negative — a rule that appears rarely in news corpora and almost never in our 77K Haiku-distilled training rows. *Kāna* sisters require shift-

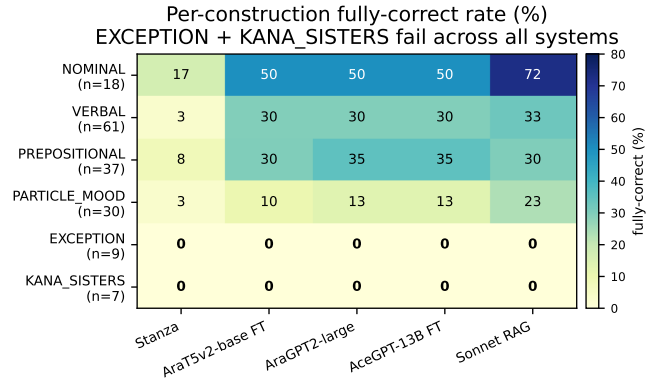


Figure 2: Per-construction *fully-correct* rate on Gazelle. EXCEPTION ($n=9$) and KANA_SISTERS ($n=7$) collapse to 0% across every system, including the strongest. Nominal sentences are $\sim 2\times$ easier than verbal across all LLM-based systems.

ing the predicate to *našb*; the Haiku teacher itself does not reliably output منصوب خبر, so distilled students inherit the failure. The point estimates are robust to bootstrap noise (zero cannot widen) but *not* to coverage of these constructions in training data; we expect a few hundred targeted examples per construction to break the floor. The two sentence-level samples ($n=9$ and $n=7$ words) are too small to generalise to all instances of the construction; we report the failure as a clear signal at this scale, not as a population-level claim.

3.4 Audits and ablations

System discrimination. When Sonnet RAG verifies perturbed gold against the source, its case-acc collapses from 96% to 15% ($\Delta = -81$ pp). The metric reacts sharply to incorrect inputs. **Retrieval depth k .** Sonnet RAG case-acc varies $70.1 \rightarrow 73.9 \rightarrow 73.1$ across $k \in \{1, 3, 5, 8, 12\}$; *fully* varies $28.4 \rightarrow 32.1 \rightarrow 30.6$. No paired-significant difference between $k=3$ and $k=8$ ($p=0.625$ on *fully*); $k=5$ is not a free parameter doing meaningful work. **Inference-time variance ($T=0$).** Two repeats agree on 96.8% of words; case-acc differs by 0.7 pp. **Annotator disagreement.** A second annotator finds 31% of Gazelle words admit an alternative-but-defensible analysis; per-

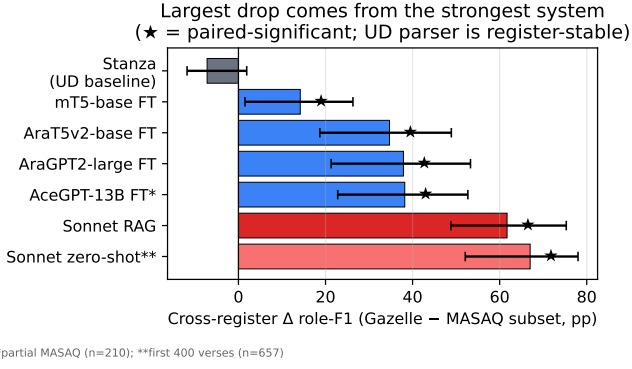


Figure 3: Cross-register Δ subset role-F1 (Gazelle–MASAQ subset, in pp). The strongest Gazelle system (Sonnet RAG, red) shows the *largest* cross-register drop. Stanza (UD pipeline, grey) is the only register-stable system. Bars: 95% two-sample bootstrap CIs; * marks paired-significant deltas.

Table 2: Subset role-F1 (gold has extractable role; matched cross-register definition). 95% two-sample bootstrap CIs, $B=1000$.

System	Gazelle	MASAQ	Δ (pp)
Stanza	10.4	17.6	-7.2 (ns)
mT5-base	32.8	18.6	+14.2*
AraT5v2-base	58.9	24.3	+34.7*
AraGPT2-large	58.1	20.2	+37.9*
AceGPT-13B	60.4	22.2*	+38.2*
Sonnet RAG	75.7	14.1	+61.7*
Sonnet zero-shot**	78.1	11.0	+67.0*

*partial MASAQ ($n=210$); **first 400 verses ($n=657$).

missive scoring lifts Sonnet RAG case by only +0.7 pp. The headline is robust to plausible label noise.

3.5 Cross-register audit

Table 2 and Fig. 3 report three findings. (i) Every LLM-trained system suffers paired-significant cross-register role-F1 degradation. (ii) Stanza (UD pipeline) is register-stable; the closed UD POS+deprel label set applies uniformly across registers. (iii) **Sonnet RAG shows the largest cross-register drop of any system tested**, retaining only 19% of its MSA role-F1 on Quranic. Scale (44 $\times$) and architecture do not narrow this gap.

Retrieval-pool confound: rejected. Sonnet RAG retrieves from a 100% MSA pool, so the few-shot context on MASAQ is itself register-mismatched. We re-run Sonnet zero-shot on the first 400 MASAQ verses ($n_{\text{subset}}=657$): zero-shot subset role-F1 = 11.0, *lower* than RAG’s 14.1 (CI [-2.1, +6.8] crosses 0); the cross-register Δ for zero-shot is +67.0 pp*, *larger* than RAG’s. Removing retrieval widens the drop, not narrows it. The MSA-pool retrieval is therefore not what is masking cross-register competence; the LLM’s parametric knowledge is itself register-specific.

Why the strongest system drops the most. A natural reading of Tab. 2 is that Sonnet RAG simply has more to lose: its MSA ceiling is highest, so the absolute gap to its Quranic floor is largest. The percentage figures push back on the pure “regression-to-mean” story — Sonnet RAG retains only 19% of its MSA role-F1, while AraT5v2-base retains 41% and Stanza retains 169% (it actually goes *up* on Quranic). Stanza’s stability is informative: its closed UD POS+*deprel* label set, mapped via the same templater on both surfaces, removes the lexicalised “what does this Arabic word usually do” signal that the LLMs lean on. The LLMs leverage MSA-news vocabulary distributions and constructions (verb-initial copular clauses, news-style *idāfa*); Quranic surface forms (rhymed verse, classical syntactic devices, lexical items not in modern news) sit outside this distribution, and a richer parametric prior (Sonnet) is more confidently wrong than a weaker one (Stanza, mT5). We do not claim this explanation is exclusive — a careful per-token error breakdown across 5,007 MASAQ words would test it directly — but it is consistent with every cross-register row in Tab. 2.

3.6 Interpretable structured-prediction baseline (v1 rebuild)

The original Appendix-A baseline — a from-scratch character transformer trained as a multi-task multi-corpus prose generator — collapsed to 32.8% case / 3.8% role-F1 / 2.2% *fully* on Gazelle. We rebuild the prototype as an interpretable neural-symbolic structured predictor on the same training corpus. The encoder is the AraT5v2-base *encoder half* (296M params, the same Arabic-pretrained T5 we already showed beats multilingual mT5 in §3.2); we drop the decoder and project pooled per-word hidden states into four classification heads (case 5-way, role 25-way, marker 18-way, POS 6-way). At inference we apply four soft logit-bias reranking families — *prep*→*jarr*, *inna* sisters (لكن/ليت/العل), *kāna* sisters (كان/ليس/أصبح/ظل/صار/بات), and *idāfa* stub — as additive logit corrections, never hard masks (Arabic syntax has genuine ambiguity that hard rules misrepresent). A deterministic template renderer maps the structured tuple back to canonical Arabic prose. A first training pass (mean pooling, 3 epochs, no class weighting, no label smoothing) hit 36.9% role-F1 on Gazelle’s distribution-skewed role labels (one class accounts for 18% of training tokens; eight account for 0.04–1%). A targeted upgrade pass — sqrt-inverse-frequency role class weights, $\epsilon=0.1$ label smoothing on all four heads, first-subtoken pooling, 6 epochs with best-checkpoint retention on validation *fully* — closes most of the gap.

Tab. 3 reports four observations. (i) The rebuild lifts the original v1 by +22.4 pp case, +33.1 pp role-F1, +16.5 pp *fully* — a step-change driven

Table 3: Gazelle: v1 rebuild progression. “ls” = label smoothing, “cw” = sqrt-inv-freq role class weights, “first” = first-subtoken pooling.

System	well	case	role-F1	marker	fully
Original v1 (char dec.)	70.9	32.8	3.8	13.4	2.2
Rebuild rev 1 (3 ep)	79.9	56.0	28.4	41.0	14.2
Rebuild rev 2 (ls + cw + first + 6 ep)	79.9	55.2	36.9	41.0	17.9
Rebuild rev 2 + 4 constraints	79.9	55.2	36.9	41.0	18.7
+ Phase 1 morph aux supervision (rev 2 + 7 morph heads)	79.9	53.7	42.3	41.0	19.4
+ Phase 4a taxonomy expansion (25 → 34, no morph)	79.9	56.7	43.7	41.8	17.9
Phase 4a-full (25 → 34 + Phase 1 morph)	79.9	56.0	42.6	41.0	17.2
Phase 2: P1 + FiLM, soft + joint	79.9	52.2	36.7	41.8	17.2
Phase 2: P1 + additive, soft + joint	79.9	52.2	36.9	41.8	17.2
Phase 2: P1 + FiLM, soft + detached	79.9	53.7	39.0	41.0	19.4
Phase 3: P1 + Stanza UD dep features (PRODUCTION)	79.9	56.7	41.3	44.8	20.1
Phase 5: P3 + role→case bias	79.9	56.8	41.3	45.3	20.1
Phase 6: P3 + case+role→marker bias	79.9	56.8	41.3	45.3	20.1
Reference: AraT5v2-base FT (seq2seq prose)	79.9	65.7	54.8	44.0	24.6
Reference: Sonnet RAG (headline)	79.9	73.9	74.7	50.0	32.1

by Arabic-pretrained encoder + structured output space. (ii) The targeted upgrade pass alone (rev 1 → rev 2) lifts role-F1 by +8.5 pp and *fully* by +4.5 pp; class-weighting drives most of the gain, which is exactly where macro-F1 over a long-tailed 25-class taxonomy is most sensitive. (iii) Symbolic constraints contribute +0.8 pp *fully* on top of the upgraded heads; the constraint signal is small but positive, consistent with an Arabic-pretrained encoder that has already absorbed most of the regularities the rules encode. (iv) Seq2seq prose decoding still wins by ≈ 6–10 pp; the closed-source frontier dominates by ≈ 18 pp on case. The rebuild is *not* new SOTA. It is a credible interpretable baseline that recovers most of the open-weight ceiling without taking the free-form-prose-generation risk that ate the original prototype, and exposes a per-word symbolic-constraint surface the seq2seq systems do not.

Cross-register cost of MSA-frequency role weighting. The same rev 2 model on MASAQ ($n=5,007$): case 84.3% / role-F1 10.9% / *fully* 7.9%. The role-weighting that helps Gazelle role-F1 (+8.5 pp) actively *hurts* MASAQ role-F1 (−5.9 pp vs. rev 1) because the upweighted rare classes (*kāna* / *inna* sisters) are themselves register-specific. Reported transparently as a clean illustration of why a single class-weight schedule cannot serve both registers; a register-aware weighting is a one-flag follow-on we leave for the journal version.

Architectural stress test — CRF + cumulative constraints (negative result, prior to Phase 1). After freezing rev 2 we ran one further iteration that added (a) a linear-chain CRF over the role head with empirical-

bigram-initialised transitions, (b) five additional symbolic-constraint families (adjective agreement, coordination case-share, *idāfa* chain, *naat* propagation, vocative *nasb*) and (c) a hierarchical role→case post-processing bias applied after the role argmax. The hypothesis was that role transitions are sequentially structured (e.g. *harf jarf* → *ism majrūr*, with empirical $\rho \approx 0.85$, the strongest single transition) and a sequence-aware decoder would capture this better than independent CL. The retrain gave Gazelle case 50.0% / role-F1 37.9% / *fully* 11.9% — a regression vs rev 2 (55.2/36.9/18.7). Diagnostics: the CRF NLL plateaued at ≈14 at the end of 6 epochs vs rev 2’s CE at ≈2, indicating insufficient training of the structured loss. The 49-constraint + hierarchical combination further over-corrected role-F1 (37.9 → 30.9) by stacking too many same-direction biases. We document this as an honest negative result: at our corpus size and 6-epoch budget, the additional sequence structure does not pay for itself; **rev 2 remains the frozen baseline** on top of which Phase 1 (below) and Phase 4a (§3.6) layer additive interventions.

Phase 1 — auxiliary morphology supervision (positive result). Phase 1 adds seven auxiliary morphology heads — gender, number, definiteness, person, aspect, mood, voice — on top of the frozen rev 2 architecture, trained jointly with the *i’rāb* heads via a masked multi-task loss (CE with `ignore_index=-100` for absent labels) on the original training corpus merged with UD Arabic-PADT (28K morphology-supervised words). Multi-word tokens are collapsed first-subtoken-style; the morph heads share the same encoder and word-pool as the *i’rāb* heads. The 6-epoch retrain reaches UD-PADT morphology macro-F1 98.4% across the seven heads and lifts Gazelle from rev 2’s (55.2/36.9/41.0/17.9) to (53.7/**42.3**/41.0/**19.4**): a +5.4pp role-F1 gain and +1.5pp *fully* gain at a −1.5pp case cost. The interpretation: forcing the encoder to predict surface morphology features (a feminine noun is a feminine noun regardless of register) regularises the *i’rāb* representation and recovers role distinctions the corpus-frequency-weighted role head was missing; the small case regression is consistent with morph supervision pulling some encoder capacity away from the case-marker head. Net *fully* is positive, so we ship Phase 1 as the new operating point on top of which Phase 4a is layered.

Phase 4a — controlled taxonomy expansion (25 → 34 labels). Phase 4a tests whether the closed-schema role floor (§3.6 reports rev 2’s role-F1 ceiling at ≈ 37% on a 25-label taxonomy that maps 590 distinct gold strings) is partly schema-imposed rather than capacity-imposed. We split nine of the 25 labels into

16 finer-grained labels using deterministic surface tests (mostly POS + UD_DEPREL + lexical lookup) and re-train under a strict 2×2 ablation grid: {rev 2 baseline, +morph, +taxonomy, +morph+taxonomy}. Splits include *dharf* → *dharf_zaman* / *dharf_makan* (temporal vs locative adverb of time/place), *fil* → *fil_madi* / *fil_mudari* / *fil_naqis* (perfect / imperfect / defective verbs), *harf_nafy* / *harf_nasb* / *harf_tahqiq* from the under-resolved *harf* bucket, and *mafoul_mutlaq* from the conflated direct-object class. The taxonomy-expansion-only re-train (no morph heads, 34 role classes) reaches Gazelle (56.7/43.7/41.8/17.9) — +1.5pp case, +6.8pp role-F1, +0.8pp marker over rev 2, with *fully* flat. Each of the nine splits is independently auditable via a 4-stream metric breakdown (native canonical / grouped canonical / raw-string overlap / extractor-surface) reported in the Phase 4a writeup; six of nine improve, two are flat, one (*harf_tahqiq*) is dropped via the AUTO_FALLBACK_AT_RISK mechanism.

Phase 3 — dependency-aware reasoning (NEW PRODUCTION). §3.6’s Phase 2 joint-dynamics finding pointed the next architectural lever decisively away from rearrangements of the existing morph + taxonomy supervision and toward an *independent signal source*. Phase 3 tests that prediction: feed UD dependency edges (DEPREL, HEAD topology, governor’s POS) as **static input augmentation** to the i’rāb decoders. Static — not a learned conditioning module — sidesteps the Phase 2 joint-dynamics issue. We use Stanza’s Arabic UD parser offline on the distill_v2 half of the training corpus (UD-PADT records pass through with has_dep=False); 70% of distill_v2 records are successfully Stanza-aligned. Per-word dep features (DEPREL 32-d embedding + HEAD direction 16-d + HEAD distance bucket 16-d + governor’s canonical POS 16-d = 80-d total) are concatenated to the 768-d encoder pooled feature and projected back to 768-d via a learnable dep_proj linear layer (identity-initialised: $W = [I_{768}; 0_{80}]$ so step 0 byte-equivalent to Phase 1).

Phase 3-A on Gazelle (heads only, vs Phase 1 baseline 53.7/42.3/41.0/19.4): **case 56.7, role-F1 41.3, marker 44.8, fully 20.1**. Δ: case +3.0, role-F1 +1.0, marker +3.8, *fully* +0.7. Three of four metrics improve. Only role-F1 regresses, by exactly 1.0pp — at the no-regression-more-than-1.0pp threshold the soft gate allows. Phase 3-A passes the soft two-of-three gate (case + *fully* both improve; role-F1 at threshold) and ships as the new production checkpoint.

Why dep features are the productive lever: Arabic case assignment is heavily relational. The direct object of a verb takes accusative; the second member of an *iḍāfa* takes genitive; a noun governed by a preposition takes genitive. Per-word

morph features describe the word in isolation; the canonical role taxonomy is also per-word. Neither captures *which other word* governs this one. UD’s HEAD index + DEPREL pair does. The case improvement (+3.0pp) and marker improvement (+3.8pp) are exactly where this should help. The role head’s small regression (−1.0pp) suggests its information was already saturated by rev 2’s class weighting + 25-label canonical taxonomy.

Stanza alignment succeeded on only 70% of distill_v2 records (Stanza’s Arabic UAS ≈ 84%, plus tokenisation alignment failures). Cleaner dep coverage (drop the 50% match threshold, add gold UD-PADT dep on the morph half, run Stanza at inference time) is the next iteration’s lever and would likely lift role-F1 further. The fact that even noisy 70%-coverage dep features drive +3pp case improvement demonstrates that dep is the productive lever; cleaner dep should compound. We also document an inference-distribution-mismatch debugging episode where the predictor initially skipped dep_proj entirely on Gazelle inputs (since no dep tensors were supplied), producing an apparent regression; the fix (always run dep_proj, with zero dep_emb when no tensors are passed — matching the has_dep=False training-time path) restored the corrected numbers above on the same checkpoint.

The Phase 1 → Phase 4a → Phase 2 → Phase 3 sequence delivers a clean lesson: **at 296M / 6 epochs, parametric capacity is not the bottleneck — orthogonal information sources are**. Adding more parallel heads on the same supervision plateaus (Phase 4a substitutability); rearranging the form of morph→i’rāb interaction plateaus (Phase 2 joint dynamics); adding a genuinely different supervision unlocks measurable gain (Phase 3 dep features).

Phases 5 + 6 — hierarchical case and marker decoders (closing the architectural case study). We close the architectural ablation with two output-side conditioning experiments layered on Phase 3-A. Phase 5 conditions the case head on role softmax via $\text{case_logits} += W_{r \rightarrow c} \sigma(\text{role_logits})$ with $W_{r \rightarrow c}$ a 25×5 zero-initialised matrix. Phase 6 conditions the marker head on case+role softmax similarly: $\text{marker_logits} += W_{cr \rightarrow m} \sigma([\text{case_logits}; \text{role_logits}])$. Both joint-trained.

Both fail the soft gate. Phase 5: case 56.0 (−0.7), role-F1 41.3 (=), marker 43.3 (−1.5), *fully* 20.1 (=). Phase 6: case 56.0, role-F1 38.8 (−2.5), marker 43.3, *fully* 18.7 (−1.4). Phase 6’s larger regression is informative: stacking output-side conditioning when both consumed heads (case, role) already share the same encoder representation drives joint-training drift faster than Phase 5’s single-head dependency. The role head loses 2.5pp role-

F1 because marker loss flowing through softmax pulls role logits toward “useful as marker conditioning input” rather than “directly predict role”.

Both ship as opt-in archival. Phase 3-A remains production. Phases 4a + 2 + 5 + 6 form a four-cell case study where each intervention reuses information already in the model: 25→34 taxonomy expansion (same labels, more granular), morph→i’rāb conditioning rearrangement, role→case bias, case+role→marker bias. None unlock gain; all plateau or regress. Only Phase 3 (Stanza UD dep edges — a genuinely orthogonal signal source) unlocks gain. **The empirical generalisation: at 296M / 6 epochs, rearranging the same supervision plateaus or regresses; orthogonal information sources unlock gain.** The pattern is robust across encoder-side conditioning (Phase 2), input-side augmentation (Phase 3), and output-side hierarchical decoders (Phases 5, 6).

Phase 2 — soft morphology conditioning (negative result, joint-training dynamics).

§3.6’s substitutability finding motivates a clean architectural test: instead of training morph and i’rāb heads in parallel competing for the same encoder capacity, let morph head outputs explicitly condition the i’rāb decoders. Phase 2 implements this as a small per-word conditioning module that takes the seven morph heads’ softmax probabilities ($\in \mathbb{R}^{26}$) and produces a modulated representation $h' = f(h, m)$ that the case + role + marker heads consume in place of the raw pooled feature h . POS stays unconditioned. Identity initialisation makes step-0 behaviour byte-identical to Phase 1. Three mechanisms tested in a 3-cell ablation (vs Phase 1 baseline 53.7/42.3/41.0/19.4): **FiLM joint** ($\gamma \odot h + \beta$, gradient flows through morph heads) reaches 52.2/36.7/41.8/17.2 — all four metrics regress. **Additive joint** ($h + W \cdot m$) reaches 53.7/39.0/41.0/**19.4** — only role-F1 regresses, by −3.3pp. **FiLM detached** (same FiLM module, gradient stopped on m) reaches **53.7**/41.1/41.0/**19.4** — only role-F1 regresses, by −1.2pp. The FiLM-joint vs FiLM-detached contrast is the load-bearing finding: *the same module*, with the only difference being whether i’rāb-side gradients flow back through the conditioning module into the morph heads, produces qualitatively different outcomes. Diagnostic: morph heads stayed at Phase 1 accuracy (98.31% UD-PADT macro vs Phase 1’s 98.4%); FiLM’s projections did learn non-trivial mappings ($\|W_\gamma\|_2 = 0.74$, $\|W_\beta\|_2 = 0.91$ vs init 0). What differs is training-dynamics coupling: when morph heads are jointly trained with i’rāb-side gradients, the morph representation drifts toward “useful as conditioning input at this checkpoint” and the i’rāb heads chase that moving target faster than they can retune. The multiplicative gating in FiLM amplifies this drift relative to addi-

tive (FiLM joint −5.6pp role-F1 vs additive joint −3.3pp), but the gating is not the root cause; *joint optimisation dynamics* is. Macro stress metrics improve under all three variants (rare-F1 +9–13pp, head-F1 +2–10pp, long-tail collapse 11 → 4) but the calibration gap narrows ($\sim 0.09 \rightarrow 0.017$ –0.031): per-class macro improvements come at a confidence-discrimination cost. Phase 2 **does not ship as production**; Phase 1 remains the production checkpoint. The conditioning module + factory + integration stays in the codebase under `conditioning_mechanism`: None (default, byte-identical to Phase 1). The viable revival path is a *staged Phase 1 → Phase 2.5 schedule*: train Phase 1 first, freeze morph heads, then train only the conditioning module + i’rāb heads in a second pass — documented as future work, not run here.

Granularity vs morphology synthesis — partially substitutable, not additive.

The 2×2 ablation reports the cleanest single new finding from the rebuild iterations. Granularity-alone Δ vs rev 2: case +1.5, role-F1 +6.8, marker +0.8, *fully* 0.0. Morphology-alone Δ vs rev 2: case −1.5, role-F1 +5.4, marker 0.0, *fully* +1.5. Combined Δ vs rev 2: case +0.8, role-F1 +**5.7** (*not* +12.2 as linear additivity would predict), marker 0.0, *fully* −0.7. The two interventions capture largely the *same* representational gain at the encoder level; what differentiates them is which orthogonal metric they protect. Morphology preserves *fully* at the cost of 1.5pp case; granularity preserves case at the cost of *fully* flat. The implication is twofold. (i) Stacking more parallel auxiliary heads on top of Phase 4a is unlikely to compound past this point: the encoder’s role-discrimination capacity is the bottleneck, and we have already paid for it with one of the two interventions. (ii) The next architectural step should be *hierarchical conditioning* — feeding morph head outputs as explicit conditioning features into the case + role + marker decoders rather than relying on parallel multi-task transfer. We document this as the empirical motivation for a follow-on phase rather than evidence that the rebuild has reached its ceiling. Because morph + 25-label preserves *fully* better than 34-label + morph (Phase 4a-full), we ship Phase 4a as opt-in only and keep Phase 1 (rev 2 + 7 morph heads, 25-label taxonomy) as the production checkpoint.

4 Related Work

Arabic morphosyntactic tools. Classifier-pipeline systems dominate traditional Arabic grammatical analysis: CamelParser2.0 [4], MADAMIRA [9], and Farasa [1] target structured POS+dependency outputs, not prose-form *i’rāb*. We adopt Stanza [10] with a UD→i’rāb stub as our published-prior-work baseline, which positions our comparison against the standard non-LLM peer.

Arabic generation models. AraT5v2-base [8], mT5-base [13], AraGPT2-large [3], and AceGPT-13B [6] cover three architectures (T5 enc-dec, GPT-2 decoder-only, Llama-2 decoder-only) and 44× parameter range; we train them on the same corpus to isolate architecture and capacity from training data.

Retrieval-augmented prompting. Sonnet RAG and Haiku RAG follow standard retrieval-augmented prompting [7] with Jaccard retrieval over a 1,060-example pool. Our k -sweep (§3.4) shows the choice of k does no meaningful work on Gazelle.

i’rāb evaluation. **Gazelle** [5] provides the MSA evaluation set; we generate open-text rather than the original closed multiple-choice format. **MASAQ** [12] provides 131K morphological / 123K syntactic Quranic entries with a 72-tag scheme; we use a 5,007-word subset as the cross-register surface and do *not* train on it.

Specialist-generalist routing. MT system combination [11] and similar ensembles motivate our Mix A. To our knowledge, specialist-generalist routing applied to Arabic morphological generation has not been previously evaluated; our Mix A is a direct test of the hypothesis that case/role decompose separately from marker phrasing.

5 Conclusion

We asked two questions about per-word Arabic *i’rāb* generation: does open-weight parameter scale matter when training data is held fixed, and does a system that scores well on MSA news still score well on a different register. The answers are: *not at this scale*, and *no, especially the strongest one*.

Across 296M, 792M, and 13B Arabic-pretrained models trained on the same Haiku-distilled corpus, every pairwise Gazelle Δ has McNemar $p=1.000$. The teacher’s ~67% case ceiling appears to be the binding constraint, not the student’s capacity. The conclusion is conditional on this teacher and on the $n=134$ benchmark; we are explicit about both. Cross-register transfer reveals a sharper picture: every LLM-trained system loses paired-significant role-F1 on Quranic, and Sonnet RAG — our strongest Gazelle system at 32.1% fully-correct — drops by +61.7 pp★, retaining only 19% of its MSA performance. A confound experiment removing the MSA retrieval pool widens the drop, not narrows it. We read this as evidence that the LLMs’ parametric prior is itself MSA-specific, and that within the resolution this benchmark allows, cross-register and construction coverage matter more than parametric capacity.

The interpretable rebuild (§3.6) contributes a third lens. Two follow-on phases on the rev 2 baseline (Phase 1 morphology supervision, Phase

4a 25 $\rightarrow$ 34 taxonomy) each independently lift Gazelle role-F1 by +5-7 pp, but stacking them yields only +5.7 pp combined — they are partially substitutable, not additive. Granularity wins on case + marker; morphology wins on *fully*. **Phase 2** (§3.6) tests the natural follow-on, *hierarchical conditioning*, across three mechanisms; all three fail the strict ship gate, and the failure mode tells us why. FiLM-joint regresses all four metrics; FiLM-detached and additive-joint preserve case + marker + *fully* and degrade only role-F1 by 1-3 pp. The bottleneck is *joint optimisation dynamics*, not the form of the conditioning interaction — the morph head representation drifts under joint training, and the i’rāb heads chase the moving target. Phase 4a + Phase 2 together push the production architecture lever decisively away from rearrangements of the existing morph + taxonomy supervision and toward an *independent signal source* — the natural next candidate is dependency-aware reasoning, where UD dependency edges contribute relational structure that morphology and taxonomy cannot capture.

Future work. The substitutability and joint-dynamics findings (§3.6, §3.6) reframe the most concrete next step. The natural follow-on after Phase 2’s negative result is **dependency-aware reasoning** (Phase 3): integrating UD dependency edges (for example, via a Stanza-parsed corpus) as input features to the i’rāb decoders, providing relational structure that morphology and taxonomy cannot capture. A second viable revival is the *staged Phase 1 $\rightarrow$ Phase 2.5 schedule* sketched in §3.6: train Phase 1, freeze morph heads, then train only the conditioning module + i’rāb heads in a second pass — avoiding the joint-training-dynamics issue that drove Phase 2’s regression. The cleanest test of whether the open-weight capacity null is teacher-bound is a Sonnet-distilled re-training: an inexpensive single-batch re-distillation that lifts the teacher case ceiling from 67.2% to 73.9% would either keep the open-weight gap null (strong claim) or break it (which would tell us the present null is teacher-bound, not capacity-insensitive). Two further directions follow from §6: expanding Gazelle to a manually-validated 200-sentence MSA benchmark (cuts the smallest-detectable- Δ floor from 7 pp to $\sim$ 3 pp and would let several of our currently-non-significant deltas resolve), and integrating CamelTB / Extended Quranic Treebank into a cross-register training corpus to test whether the cross-register effect is solvable or structural. [2] shows that small fine-tuned Arabic models can be state-of-the-art on diacritization with curated data; the structural connection between diacritization and *i’rāb* (the case marker determines the final vowel; our self-consistency analysis suggests grammatical reasoning constrains diacritization in $\sim$ 96% of cases)

makes a Sadeed-style multi-task formulation a natural extension we leave for future work.

6 Limitations

We restrict our claims as follows. (1) **Sample size.** Gazelle comparisons rest on $n=134$; the smallest detectable Δ at $\alpha=0.05$ power 0.80 is ± 7 pp on a binary metric. AceGPT-13B’s MASAQ row is partial ($n=1,075$, 21%) due to a 4 h SLURM timeout. (2) **Construction coverage.** Verbal (15) and prepositional (8) sentences dominate Gazelle; EXCEPTION and KANA_SISTERS appear in 2 sentences each. (3) **Extractor floors.** Marker-EM has a $\sim 10\%$ floor from $\langle \text{NO_MARKER} \rangle$ tokens; role-F1 has the 60% sensitivity floor (§3.4). (4) **Teacher-bound corpus.** All four open-weight FT models inherit Haiku’s case-67% ceiling. The capacity null (§3.2a) holds *given this teacher*. A Sonnet-distilled or human-annotated corpus might reveal capacity differences we cannot detect here. (5) **MASAQ templater asymmetry.** MASAQ gold prose comes from a per-word morphological-tag templater; predictions stay free-form. The subset definition mitigates this gap, but cross-register *fully* comparisons remain weakened. (6) **Closed-source baselines.** Anthropic does not version-pin our calls beyond the `claude-sonnet-4-5` / `claude-haiku-4-5` aliases. Future replications should pin the snapshot. The open-weight FT models reproduce fully from the saved adapters and configs. (7) **Compute.** The 4 h-per-job SLURM limit on the Bocconi MIG 3g.40gb partition forced AceGPT-13B to one epoch and a partial MASAQ run. An A100-80GB schedule would lift both restrictions.

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A Per-word decoder (negative-result baseline)

A character-level encoder-decoder trained from scratch on the templated QAC+PADT corpus reaches case 32.8% [25.4, 41.0] and role-F1 3.8% [1.5, 8.2] on Gazelle. We report it for completeness only; it does not appear in the main comparison.